Functorial Resolution Geometry:

Observable Quotients, Admissible Sectors, and Latent-Realization Underdetermination

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Abstract

We define a measurement-side Resolution Geometry (RG) whose primary object is the statistical observational quotient rather than a selected latent representative. An admissible experiment bundles a statistical experiment with a resolution metric, threshold family, nuisance relation, stability and conditioning rules, and transport protocol. We prove that every experiment-internal construction factors uniquely through equality-of-law equivalence classes; construct generalized-Fisher resolved and null sectors covariantly on the quotient; prove naturality under experiment isomorphisms, information monotonicity under parameter-independent garblings, local stratification at admissibility boundaries, and gluing under transport cocycles; and establish latent-realization underdetermination. Synthetic attacks include latent dimensions up to 120, nonlinear and chaotic hidden augmentations, coordinate condition numbers near 6×10^{11}, protocol changes, noisy coarse-grainings, threshold crossings, observable-memory adversaries, and continuous-time stochastic hidden augmentations. The final universality attack shows that the observational quotient, not a thresholded resolved sector, possesses the unconditional universal factorization property. Thus $W_{\mathrm{obs|adm}}$ is canonical relative to a bundled admissibility protocol but is not proved initial among all conceivable observable constructions. The results establish operational representative-independence, not the nonexistence of latent manifold points.

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1 Introduction

Inverse problems frequently admit many latent representatives with the same observation law. The elementary identity $A(x+n) = A(x)$ for $n \in \ker A$ is the deterministic linear instance of a more general statistical equivalence: two parameters are observationally identical when they induce the same probability law. The mathematical question addressed here is not whether latent points exist, but which structures can be attributed to a declared experiment without importing information from a representative, a prior, a regularizer, or a separate protocol.

The central ordering is

$$\text{experiment} \longrightarrow \text{observational quotient} \longrightarrow W_{\text{obs|adm}} \longrightarrow (R, N, F, T, \Sigma).$$

The quotient is universal for all experiment-internal constructions. The remaining geometry is canonical only after an admissibility protocol has been bundled with the experiment. This distinction is forced by protocol-dependence and coordinate-adversary attacks.

Main results. We prove: (i) equality-of-law quotient factorization; (ii) naturality of covariant generalized-Fisher sectors; (iii) identity and composition laws on experiment isomorphisms; (iv) Fisher monotonicity under garblings; (v) local constancy away from admissibility boundaries; (vi) latent-realization underdetermination; and (vii) observable-atlas gluing under cocycles. We also prove a negative result: $W_{\text{obs|adm}}$ is not characterized as an initial object among all admissible constructions by the present axioms.

2 Admissible statistical experiments

Definition 2.1 (Statistical experiment). *A statistical experiment is a triple*

$$\mathcal{E} = (\Theta, \mathcal{O}, \{P_\theta\}_{\theta \in \Theta}),$$

where Θ is a parameter space, $\mathcal{O}$ a measurable observation space, and P_θ a probability law on $\mathcal{O}$.

Definition 2.2 (Observational equivalence and quotient). *Define*

$$\theta \sim_{\mathcal{E}} \theta' \quad \Longleftrightarrow \quad P_\theta = P_{\theta'}.$$

The observational quotient is $Q_{\mathcal{E}} = \Theta/\sim_{\mathcal{E}}$ with canonical projection $\pi_{\mathcal{E}} : \Theta \to Q_{\mathcal{E}}$.

Definition 2.3 (Admissibility protocol). *An admissibility protocol is a bundle*

$$\Gamma = (G, \tau, \mathcal{N}, \mathcal{S}, \mathcal{K}, \mathcal{T}),$$

where G is a parameter-comparison metric or scale structure, τ a threshold family, $\mathcal{N}$ a nuisance relation, $\mathcal{S}$ a stability rule, $\mathcal{K}$ conditioning and coarse-graining rules, and $\mathcal{T}$ a transport/gluing protocol. An admissible experiment is a pair $(\mathcal{E}, \Gamma)$.

Definition 2.4 (Morphisms). *An isomorphism of admissible experiments consists of bijections on quotient parameters and observations that identify the full conditional laws and transport Γ covariantly. A garbling morphism is a parameter-independent Markov kernel on observations, together with the induced protocol map. Isomorphisms and garblings are distinguished: the former preserve information, while the latter may destroy it.*

3 Universal quotient and factorization

Theorem 3.1 (Observable factorization)**.** *Let Z be a set and let $\Phi : \Theta \to Z$ satisfy*

$$P_\theta = P_{\theta'} \implies \Phi(\theta) = \Phi(\theta').$$

Then there exists a unique $\widetilde{\Phi} : Q_{\mathcal{E}} \to Z$ such that

$$\Phi = \widetilde{\Phi} \circ \pi_{\mathcal{E}}.$$

Proof. Define $\widetilde{\Phi}([\theta]) = \Phi(\theta)$. Constancy on equivalence classes makes this well-defined. Existence follows immediately. If $\Psi \circ \pi_{\mathcal{E}} = \Phi$, surjectivity of $\pi_{\mathcal{E}}$ implies $\Psi([\theta]) = \Phi(\theta) = \widetilde{\Phi}([\theta])$, proving uniqueness. □

Corollary 3.2 (Representative-independence)**.** *No conclusion derived solely from $\mathcal{E}$ may depend on a chosen representative of $[\theta] \in Q_{\mathcal{E}}$.*

Corollary 3.3 (Deterministic kernel quotient)**.** *For a deterministic linear map $A : X \to O$, observational equivalence is $x \sim x'$ iff $x - x' \in \ker A$, and $Q_{\mathcal{E}} \cong X / \ker A$.*

Remark 3.4. The theorem establishes operational non-identifiability. It does not imply that representatives do not exist or that another experiment cannot refine the equivalence relation.

4 Admissible observation geometry

Assume a regular finite-dimensional local model with Jacobian J, observation covariance $C \succ 0$, and protocol metric $G \succ 0$. Define

$$F = J^\top C^{-1} J.$$

The dimensionless resolution spectrum is the generalized spectrum

$$F v_i = \lambda_i G v_i.$$

Definition 4.1 (Resolved and null sectors)**.** *For threshold τ, define*

$$R_\Gamma = \operatorname{span}\{v_i : \lambda_i > \tau\}, \qquad N_\Gamma = \operatorname{span}\{v_i : \lambda_i \le \tau\}.$$

After nuisance closure, stability, conditioning, and transport gates, the retained sector is denoted $W_{\mathrm{obs|adm}}$ and its effective null complement by N_{adm}.

Theorem 4.2 (Naturality under experiment isomorphism)**.** *Let $\theta' = S\theta$ and $y' = Uy$, with S, U invertible, and transport*

$$J' = UJS^{-1}, \quad C' = UCU^\top, \quad G' = S^{-\top} G S^{-1}.$$

Then the generalized spectrum is invariant and

$$R'_\Gamma = S R_\Gamma, \qquad P' = SPS^{-1},$$

where P is the G-orthogonal resolved projector.

Proof. $F' = S^{-\top} F S^{-1}$. If $Fv = \lambda G v$, then with $v' = Sv$ one has $F'v' = \lambda G' v'$. Threshold membership is therefore preserved. The projector identity follows from transport of the G-orthonormal basis. □

Corollary 4.3 (Functor laws on isomorphisms)**.** *The assignment preserves identities and composition. Hence it defines a functor on the isomorphism subgroupoid of* **AdmExp**.

Theorem 4.4 (Information monotonicity). *If Z is obtained from Y through a parameter-independent Markov kernel, then*

$$F_Z \preceq F_Y.$$

Thus garbling cannot create Fisher information, although it may remove or rotate retained directions according to the bundled protocol.

Proof. The score of Z is the conditional expectation of the score of Y given Z. Conditional Jensen gives $a^\top F_Z a \le a^\top F_Y a$ for every a. □

Theorem 4.5 (Admissible stratification). *For a continuous family $(\mathcal{E}_g, \Gamma_g)$, the dimension of $W_{\mathrm{obs}|\mathrm{adm}}(g)$ is locally constant wherever generalized eigenvalues, stability statistics, conditioning statistics, nuisance rank, and transport bounds remain separated from their protocol boundaries. Sector changes can occur only on the corresponding discriminant set.*

Proof. Generalized eigenvalues of continuous symmetric-definite matrix pencils vary continuously. Every strict gate inequality persists locally. Therefore no direction enters or leaves until at least one boundary is reached. □

5 Transport and gluing

Definition 5.1 (Observable atlas). *An observable atlas consists of local admissible sectors W_i, bounded transports $T_{ij} : W_i \to W_j$ on overlaps, identity laws $T_{ii} = \mathrm{id}$, and cocycle laws*

$$T_{jk}T_{ij} = T_{ik}$$

on triple overlaps.

Proposition 5.2 (Gluing criterion). *Pairwise bounded transports define a consistent global atlas only when the identity and cocycle laws hold. Failure of the cocycle produces path-dependent comparison and obstructs gluing.*

Proof. If the cocycle holds, coordinate transfer is independent of the chosen two-step refinement on triple overlaps. Conversely, if $T_{jk}T_{ij} \ne T_{ik}$, the same local element receives two different images, so no consistent identification exists. □

6 Latent-realization underdetermination

Theorem 6.1 (Latent realization underdetermination). *Suppose latent models X_1 and X_2 induce isomorphic quotient statistical experiments and the same transported protocol Γ. Then their admissible observation geometries are naturally isomorphic. No inference internal to the experiment uniquely determines latent dimension, hidden coordinates, or a representative within an observational class.*

Proof. Both experiments factor through isomorphic quotients by the observable factorization theorem. Naturality transports the generalized-Fisher geometry and every covariant admissibility gate. Therefore the outputs are naturally isomorphic. □

Corollary 6.2 (Continuous-time version). *Let two stochastic latent processes induce the same law on observed path space, and let the protocol depend only on that observed law. Then every experiment-internal path functional, Fisher construction, and admissible sector factors through the observed path-law quotient. Hidden stochastic coordinates annihilated by the observation process remain underdetermined.*

Remark 6.3 (Memory and holonomy). A path integral, memory statistic, or holonomy is admissible only if it is determined by the observed path law. A functional of hidden trajectories does not factor through the experiment and requires an enlarged experiment or an explicit prior assumption.

7 Universality boundary

Theorem 7.1 (Universal property of the observational quotient)**.** *The quotient projection $\pi_{\mathcal{E}} : \Theta \to Q_{\mathcal{E}}$ is universal among maps constant on equality-of-law classes: every such map factors uniquely through $\pi_{\mathcal{E}}$.*

Proof. This is the observable factorization theorem restated as the universal property of a quotient/coequalizer in the relevant concrete category. □

Proposition 7.2 (Failure of unrestricted initiality for $W_{\mathrm{obs|adm}}$)**.** *The current axioms do not make a thresholded sector $W_{\mathrm{obs|adm}}$ initial among all admissible observable constructions.*

Proof. Different bundled thresholds produce different sectors from the same quotient experiment. Even with a fixed object, covariance alone permits multiple natural rescalings or target constructions unless normalization and target morphisms are further restricted. Hence a unique natural transformation from $W_{\mathrm{obs|adm}}$ to every admissible construction is not forced. □

Corollary 7.3 (Correct canonical claim)**.** *The observational quotient is unconditionally universal for experiment-internal constructions. The RG geometry is a canonical, natural construction relative to $(\mathcal{E}, \Gamma)$, but is not presently characterized as the unique construction satisfying a minimal axiom set.*

8 Numerical attacks

All synthetic calculations used seed `20260803`. Earlier quotient attacks used seed `20260802`. Complete machine-readable outputs are stored in `rg_final_universality_attack.json`, `rg_admexp_category_lock_battery.json`, and `rg_observable_quotient_invariance_battery.json`.

8.1 Category-lock summary

Table 1: Selected category-lock results.

Test	Result
Base generalized spectrum	$(2.429, 4.639, 5.193, 5.741, 12.468)$
Resolved dimension at $\tau = 4.916$	3
Composition experiment error	3.40×10^{-15}
Composition sector angle	9.03×10^{-14} degrees
Largest tested raw Fisher condition number	5.99×10^{11}
Resolved dimension after covariance scaling $C \mapsto 2.5C$	$3 \to 1$
Threshold sweep dimensions	$5, 4, 3, 1, 0$

The covariant generalized sector retained dimension three under coordinate scale spans through 10^3; its transported projector error remained 1.56×10^{-6} despite raw Fisher conditioning near 6×10^{11}. Three latent realizations of dimensions $8, 50, 120$ induced quotient-Fisher relative errors below 5.3×10^{-16}. Noisy coarse-graining gave positive minimum eigenvalues of $F_Y - F_Z$ in all four tests.

8.2 Final universality attack

The finite quotient test used 60 latent states partitioned into 12 observational classes. A randomly generated class-constant construction factored with zero numerical error and zero within-class variation. A representative-dependent adversary produced nonzero within-class variation and therefore failed admissibility.

The initiality attack compared two threshold sectors of one spectrum. Multiple scalar natural transformations remained possible absent normalization axioms, so uniqueness was not forced. This falsifies the unrestricted universal claim for $W_{\mathrm{obs|adm}}$ while preserving universality of $Q_{\mathcal{E}}$.

A history attack used one observed trajectory with two distinct hidden-memory processes. Observable memory was identical, while hidden terminal memory differed. Thus memory does not defeat factorization; it moves the correct quotient from point laws to observed path laws.

A continuous-time Ornstein–Uhlenbeck observation was augmented by hidden stochastic systems of dimensions $1, 8, 40$ and different relaxation rates. The observed state path and observed-path Fisher proxy were unchanged. This extends latent-realization underdetermination to the tested continuous-time stochastic class.

Finally, exact transports satisfied the cocycle to numerical zero, whereas a 10^{-2} perturbation produced a nonzero cocycle defect. Pairwise transport is therefore insufficient without gluing compatibility.

9 Relation to established mathematics and scope of novelty

The equality-of-law quotient and its universal property belong to classical quotient and identifiability mathematics. Sufficiency formalizes lossless reduction of an experiment, while Blackwell comparison formalizes information loss under randomization. Fisher monotonicity and coordinate covariance are likewise established ingredients. RG does not claim invention of these components.

The proposed contribution is their closure into a protocol-explicit observation geometry: the quotient is followed by covariant resolved/null sectors, stability and conditioning gates, nuisance closure, stratification, and gluable transport. The novelty claim is therefore architectural and theorem-compositional. A stronger characterization analogous to a uniqueness theorem for all natural information metrics is not proved here.

10 Non-claims and falsification criteria

Non-claim 10.1 (No nonexistence of latent points). The theory does not prove that smooth manifolds lack points or that latent representatives do not exist. It proves that an experiment-internal conclusion cannot distinguish representatives identified by the observation law.

Non-claim 10.2 (No protocol-independent uniqueness). Changing G, thresholds, nuisance rules, or transport conventions may change the output geometry. Uniqueness is relative to the bundled object.

Non-claim 10.3 (No unrestricted initiality). $W_{\mathrm{obs|adm}}$ is not shown initial among every admissible functor. The universal object presently proved is the observational quotient.

Non-claim 10.4 (No physical ontology from factorization). Latent underdetermination does not imply that observationally equivalent models are ontologically identical.

The framework is falsified or requires revision if a purported experiment-internal quantity varies within an equality-of-law class; if the generalized construction fails naturality under a correctly transported experiment isomorphism; if a declared garbling creates Fisher information under a regular correctly specified model; or if proposed global transports violate cocycle compatibility.

11 Conclusion

The final attack locates the exact universal core of Resolution Geometry. Equality of observation laws induces a quotient through which every experiment-internal construction factors uniquely.

A bundled admissibility protocol then determines a natural stratified observation geometry with covariant generalized-Fisher sectors, information-monotone garblings, and gluable transports. This geometry is independent of latent realization and extends to observed-path statistical experiments. The thresholded sector $W_{\mathrm{obs|adm}}$ is canonical relative to the bundle but is not proved initial among all possible constructions. The strongest justified statement is therefore operational rather than metaphysical:

A latent point is not an experiment-internal observable object unless the experiment separates it.

Reproducibility

The final attack output is `rg_final_universality_attack.json`. Supporting outputs are `rg_admexp_category_lock_battery.json` and `rg_observable_quotient_invariance_battery.json`. Seeds, dimensions, matrices, thresholds, and all reported diagnostics are stored in those files. The final release should additionally include the executable analysis script, dependency versions, and hashes of all artifacts.

References

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